

Short README – Analytical Problem Solving (AI Lab)

Experiment Objective

The aim of this experiment is to understand how Artificial Intelligence solves problems using search algorithms, intelligent agents, and problem formulation techniques.

1. Water Jug Problem

Objective: Obtain exactly **2 gallons** of water in the 4-gallon jug using the allowed operations (fill, empty, and pour).

Concept Used: State Space Search

Outcome: The goal is achieved by performing a sequence of valid jug operations.

2. Mars Rover Agent

Objective: Study how the Mars Rover works as an intelligent agent.

Key Points:

- **Percepts:** Camera images, sensors, rock and soil data.
 - **Actions:** Move, collect samples, analyze, send data.
 - **Environment:** Dynamic and partially observable.
 - **Agent Type:** Utility-Based Learning Agent.
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3. 8-Queens Problem

Objective: Place 8 queens on a chessboard so that no two queens attack each other.

Concept Used: Constraint Satisfaction Problem (CSP)

Search Method: Backtracking / Depth-First Search (DFS)

4. OLA Cab Booking Agent

Objective: Select the most suitable cab based on the customer's preferences.

Agent Type: Goal-Based Agent

Working: The agent checks available cab types, selects one based on the user's choice, confirms the booking, and assigns a driver.

5. Uniform Cost Search (UCS)

Objective: Find the least-cost path from **S** (Start) to **G** (Goal) in a weighted graph.

Concept Used: Uniform Cost Search (UCS)

Result: UCS always expands the node with the lowest cumulative cost and finds the optimal path.
